

(D) 2 or -2

6. If  $n=10^{10}-1$ , the number of digits in  $n^3$  is

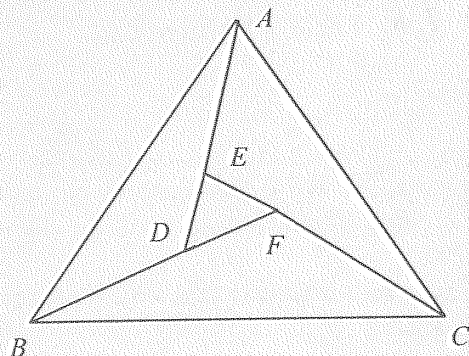
(A) 30

(B) 28

(C) 32

(D) 27

7.



In the adjoining figure  $ABC$ ,  $DEF$  are equilateral triangles.  $AB = 8\text{cm}$  and  $DE = 3\text{cm}$ . Then the possible value of  $AE + BD + CF$  is

(A) 6.9cm

(B) 7.1cm

(C) 5.2

(D) 8.3

8. Anu, Babu and Chitra have 51 balls altogether. Babu gives 7 balls to Chitra and Chitra gives 5 balls to Anu and Anu gives 4 balls to Babu. If all the three have equal number of balls then the number of balls Anu had at the start is

(A) 17

(B) 20

(C) 16

(D) 19

9. The number of ordered triples  $(a,b,c)$   $1 \leq a,b,c \leq 9$  such that  $ac = b^2 - 1$  is

(A) 9

(B) 7

(C) 14

(D) 18

10. Let ' $m$ ' be the number of perfect squares in  $1,2,3, \dots, 1000000$ . Let ' $n$ ' be the number of perfect cubes in  $1,2,3, \dots, 1000000$ . Then the value of  $\frac{n}{m}$  is

(A) 0.01

(B) 0.1

(C) 10

(D) 100

11. The measure of one of the angles of a right triangle is five times that of a second angle. Then the possibility of the second largest angle is

(A)  $72^\circ$

(B)  $75^\circ$

(C)  $72^\circ$  Or  $75^\circ$

(D) none of these

12. The difference of the squares of two consecutive natural numbers is 2008.

(A) Only one such pairs exist

(B) Infinitely many such pairs exist

(C) No such pair exists

(D) Exactly two such pass exist

13. The value of a commodity is increased by  $x\%$  first and again increased by  $x\%$ . The total increase is

(A)  $2x\%$                       (B)  $4x\%$                       (C)  $\left(2x + \frac{x^2}{100}\right)\%$                       (D)  $x^2\%$

14.  $x$  is  $1\frac{1}{6}$ th of  $3\frac{3}{4}$  and  $y$  is  $2\frac{1}{3}$ rd of  $2\frac{1}{6}$ . Then

(A)  $2x = y$                       (B)  $y < x$                       (C)  $x < y$                       (D)  $x = y$

15.  $a, b, c$  are three non-zero digits. The sum of all the numbers formed by these three digits is always divisible by

(A)  $a + b + c$                       (B) 222                      (C)  $9(a + b + c)$

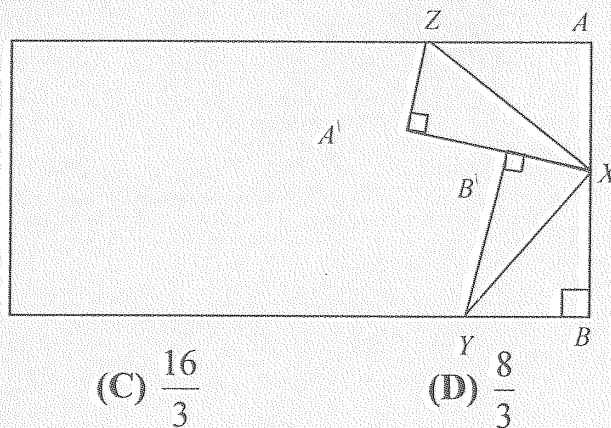
(A) only (A) is true                      (B) Only (A) and (B) are true  
(C) only (B) and (C) are true                      (D) only (A) and (C) are true

16. A leap year started on a Monday. The days appearing 53 times are

(A) Sunday and Monday                      (B) Monday and Tuesday  
(C) Tuesday and Wednesday                      (D) Saturday and Sunday

17. The number  $16^{12}$  is obtained from the number  $8^4$  by raising the smaller number to the power  $n$ . then  $n$  is

(A) 3                      (B) 4                      (C)  $\frac{16}{3}$                       (D)  $\frac{8}{3}$



18. A rectangular sheet of paper is folded so that the corners  $A, B$  go to  $A', B'$  as in the figure. Then  $\angle ZXY$  is

(A) an acute angle                      (B) an obtuse angle  
(C) a right angle                      (D) a variable angle depending on the point  $X$

19. The tens place of two three digits numbers are each 8 and both the numbers are divisible by 4. Then the difference between the biggest and the smallest such numbers is

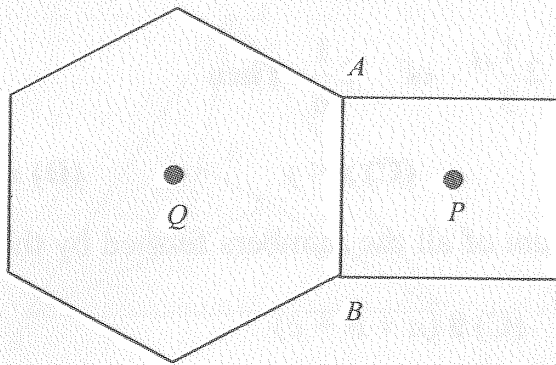
(A) 888

(B) 808

(C) 708

(D) 788

20.



(A)  $1004\sqrt{3}$

(B)  $1004(\sqrt{3}+1)$

(C)  $2008\sqrt{3}$

(D)  $1004(\sqrt{3}-1)$

On a line segment  $AB = 2008\text{cm}$  a square and a hexagon are drawn as shown in the diagram. The distance between their centres  $P, Q$ , in cm is